

Specific-Factors Model

ECON 404: Cal Poly

- Recall some of the problems we had with the Ricardian Model
 - Everyone gains without any sort of redistribution.
 - Too much specialization.
 - Labor only input.
 - Exogenous technology. Why doesn't information about production techniques transfer across countries??
- The Specific-Factors model addresses these critiques
 - Allow for more inputs than labor
 - Assume technology *could* be the same across countries

Setup of the model

- Economy can produce two goods: Manufacturing and Agriculture
- Each country has *three* factors of production: Labor (L), capital (K), and land (T for terrain)
- Manufacturing is produced using capital and labor but NOT land
- Agriculture is produced using land and labor but NOT capital
- Labor is *mobile* while land and capital are *specific* factors
- We assume diminishing marginal productivity of labor.
- The lack of mobility of factors means this model can be thought of as a “short-run” model.

Manufacturing Production

- The production function for Manufacturing is (note K is fixed)

$$Q_M = Q_M(K, L_M) \quad (1)$$

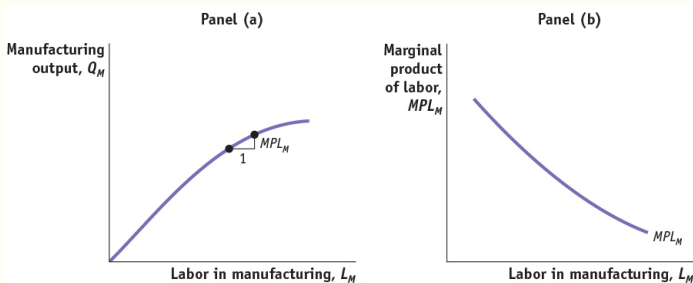


Figure 3.1
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- The production function for Agriculture is (note T is fixed)

$$Q_A = Q_A(T, L_A) \quad (2)$$

- The labor constraint is

$$L_M + L_A = L \quad (3)$$

- The production possibilities frontier is bowed out and the slope is equal to $-\frac{MPL_A}{MPL_M}$

Production Possibility Frontier (PPF)

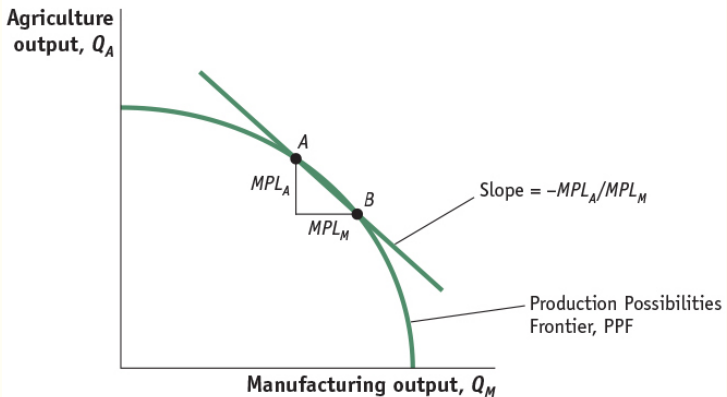


Figure 3.2

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Prices, Wages, and Labor Allocation

- How much labor will be employed in each sector?
- Looking at the supply-side of labor, labor will move to the industry that pays the most. So, with perfect competition and free labor mobility (think long run equilibrium), the wages in each sector must be equal and we need to have the revenue marginal product in each sector equal the wage:

$$w = MPL_M * P_M \quad (4)$$

$$w = MPL_A * P_A \quad (5)$$

- Rearranging these two equations, we can get

$$-\frac{MPL_A}{MPL_M} = -\frac{P_M}{P_A} \quad (6)$$

The left side of the equation is the slope of the production possibility frontier at the actual production point. The right side is the relative price of cloth.

Prices, Wages, and Labor Allocation

- What determines the relative price in Autarky??
- Utility – again try to get on the highest indifference curve given the PPF.
- Suppose there is an equal-proportional change in prices.
 - If there is no change in the relative price then there will be no change in the labor allocation and output

Suppose there is a change in relative prices.

- What happens if the price of Manufacturing increases, but the price of Agriculture remains the same?
- Going back to our wage equations:

$$w = MPL_M * P_M$$

$$w = MPL_A * P_A$$

- As the price of Manufacturing goes up, the wage rate in the Manufacturing industry must also go up.
- Labor starts to move into the Manufacturing industry out of the Agriculture industry.

Suppose there is a change in relative prices.

- However, recall that the MPL is not constant, so as labor moves into the Manufacturing industry, the MPL_M begins to decrease putting downward pressure on the wage rate in the Manufacturing industry.
- At the same time, labor is moving out of the Agriculture industry, increasing the MPL_A and thus the wage rate in the Agriculture industry.
- This happens until we are settled back into our equilibrium condition (6), however we are at a different level of production.

Suppose there is a change in relative prices

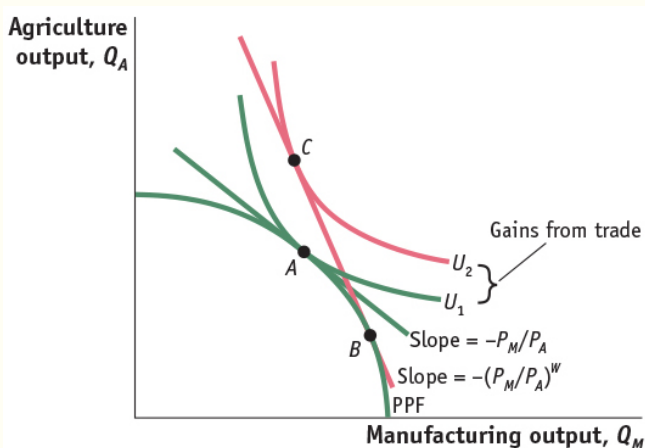


Figure 3.3

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Relative Prices and the Distribution of Income

- First note when the $p_M \uparrow$, because the MPL adjust as well, the wage rate won't increase by the same amount.
- Now let's look at what this price change implies about the incomes of three groups: workers, owners of capital, and owners of land.
 - Workers nominal wage rate increase, however it's less than the price of Manufacturing, so $\frac{w}{P_M} \downarrow$ and $\frac{w}{P_A} \uparrow$. Ambiguous effect.
 - Owners of capital are definitely better off: The real wage rate in terms of Manufacturing has fallen, so the profits of capital owners in terms of what they produce (Manufacturing) rises – i.e. the income of capital owners will rise more than proportionately with the rise in P_M .
 - Owners of land are definitely worse off. (1) The real wage in terms of food (the good they produce) rises, squeezing their income and (2) the rise in cloth price reduces the purchasing power of any given income.

- 1 The factor specific to the sector whose relative price increases is definitely better off.
- 2 The factor specific to the sector whose relative price decreases is definitely worse off.
- 3 The change in welfare for the mobile factor is ambiguous.

International Trade in the Specific Factors Model

- For trade to take place, the country must face a world relative price that differs from the autarky relative price.
- Why might the relative supply curve for the world be different from the home?
 - The other countries could have different technologies (like the Ricardian model)
 - The other countries could have the same technologies, but different resources.
- Given what we just discussed; if there is trade between two countries, the trading relative price (TOT) must be different than the countries' autarky relative price.
- Thus, even though some win and some lose in each country, the gains must outweigh the losses

Gains from Trade

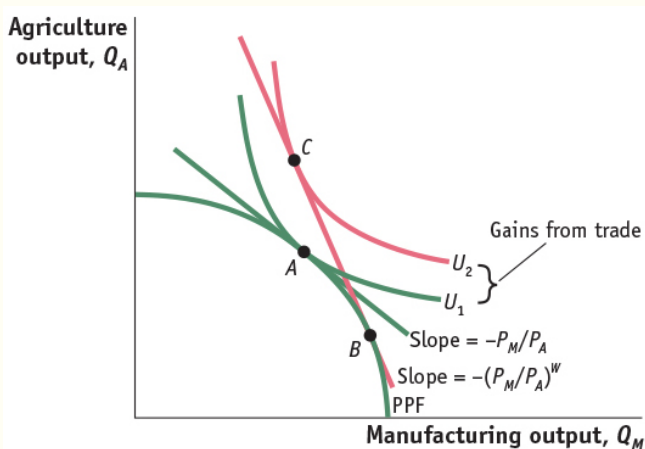


Figure 3.3

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The Political Economy of Trade: A Preliminary View

- We now have a model that highlights the fact that some win from trade liberalization (export sector) and some lose from it (import competing sector).
- There has always been this tension in society. Since there are some losers, does this mean we should not trade?? Here are three reasons why economists tend not to stress the income distribution effects of trade:
 - 1 Income distribution effects are not specific to international trade.
 - 2 It is always better to allow trade and compensate those who are hurt by it than to prohibit trade.
 - 3 The losers are typically better organized than those who gain.
- A bit more on this later, but there is the Trade Adjustment Assistance Program. For a nice discussion on it's effectiveness: <https://www.tradetalkspodcast.com/podcast/91-can-trade-adjustment-assistance-work/>

International Labor Mobility

- Just like free labor mobility within a country will equalize real wages, free labor mobility across countries will equalize real wages.
- This assumes
 - That labor is identical – though this isn't that big of an assumption because you could simply say that wages across job type will equate.
 - That labor doesn't care where it lives – think of the EU, there is theoretically free labor mobility, though people tend to stay in their own country.
- Generally, this raises world output/welfare, but of course there will be some losers if these gains aren't redistributed somehow.
- The same is true for all factors of production – though this can be difficult for other reasons.

- We now have a model that
 - ① doesn't rely on exogenous differences in technology to drive comparative advantage and thus trade.
 - ② doesn't result in *everyone* being better off without redistribution (the country as a whole is still better off with trade).
 - ③ doesn't result in full specialization if the TOT is different than autarky prices.
- This opens the door for understanding the political economy of trade (winners vs. losers)

- Next up the “Heckscher-Ohlin Model”